

Toward Third-Party Assurance of AI Systems: Design Requirements, Prototype, and Early Testing

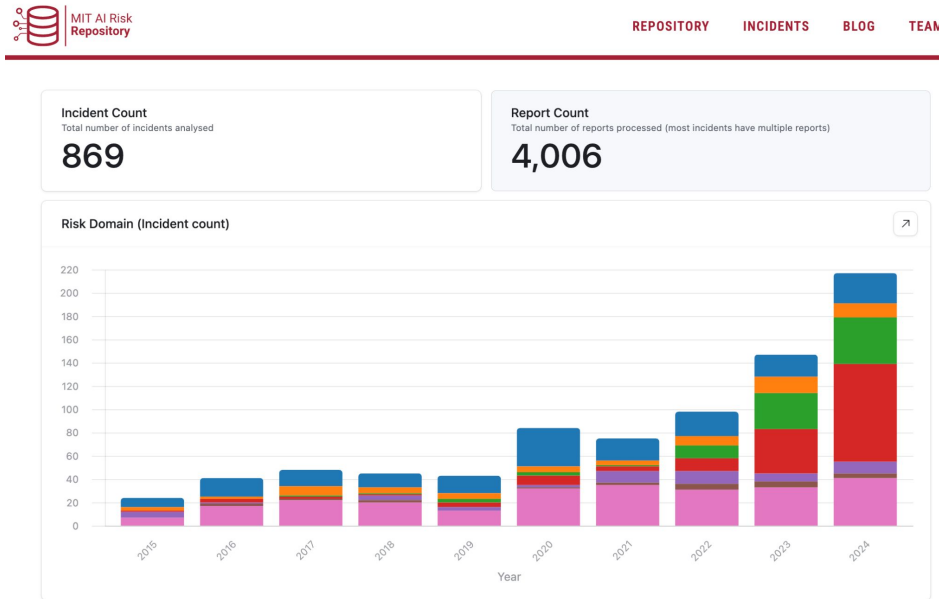
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Motivation

- AI systems are increasingly embedded in critical domains
- Need to ensure AI systems' responsible design, development, and deployment

Motivation

However, many of the existing approaches to evaluating AI systems fall short in practice.



Our Contribution

We design a **third-party AI assurance framework** that addresses the shortcomings of existing AI evaluations.

- **Third-party**: we mean that the assessment is conducted by individuals external to the team designing, developing, and deploying the AI system
- **AI assurance**: we mean an assessment of the processes behind the design, development, and deployment of an AI system and the resulting outcomes

We intend for the AI assurance framework to be **used by third-party individuals and organizations who have expertise in AI assurance, auditing, or governance.**

It should ideally occur at **multiple points throughout the AI lifecycle.**

AI Audits vs. Assurance

AI Audits

- (1) Focus on **outcomes**.
- (2) Conducted **after** an AI system has been deployed.
- (3) More **punitive**.

AI Assurance

- (1) Focus on **both outcomes** and **process**.
- (2) Conducted **throughout** the AI lifecycle.
- (3) More **proactive**.

AI Audits vs. Assurance

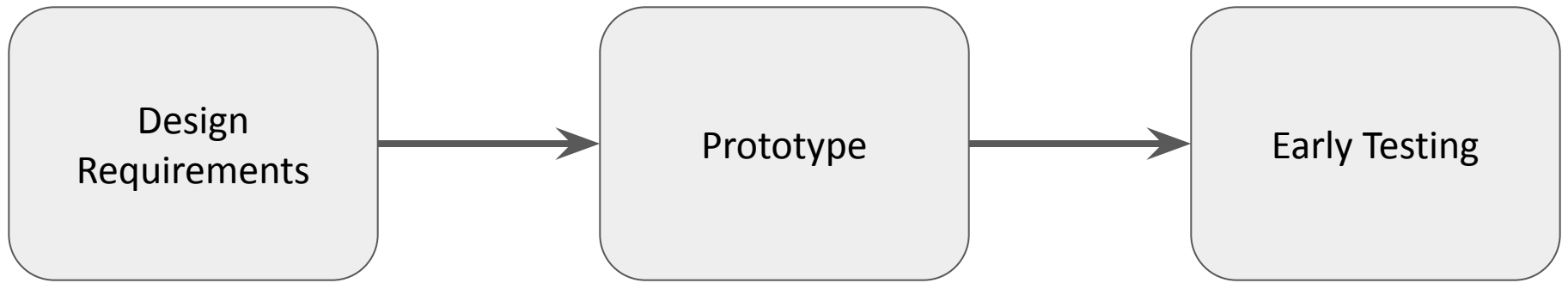
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We deliberately choose to use the term “**assurance**” to foreground a **proactive**, **improvement**-oriented, and potentially **voluntary** posture, in contrast to the more **retrospective**, **punitive**, and **compliance**-triggered connotations often associated with the term “**audit**”.

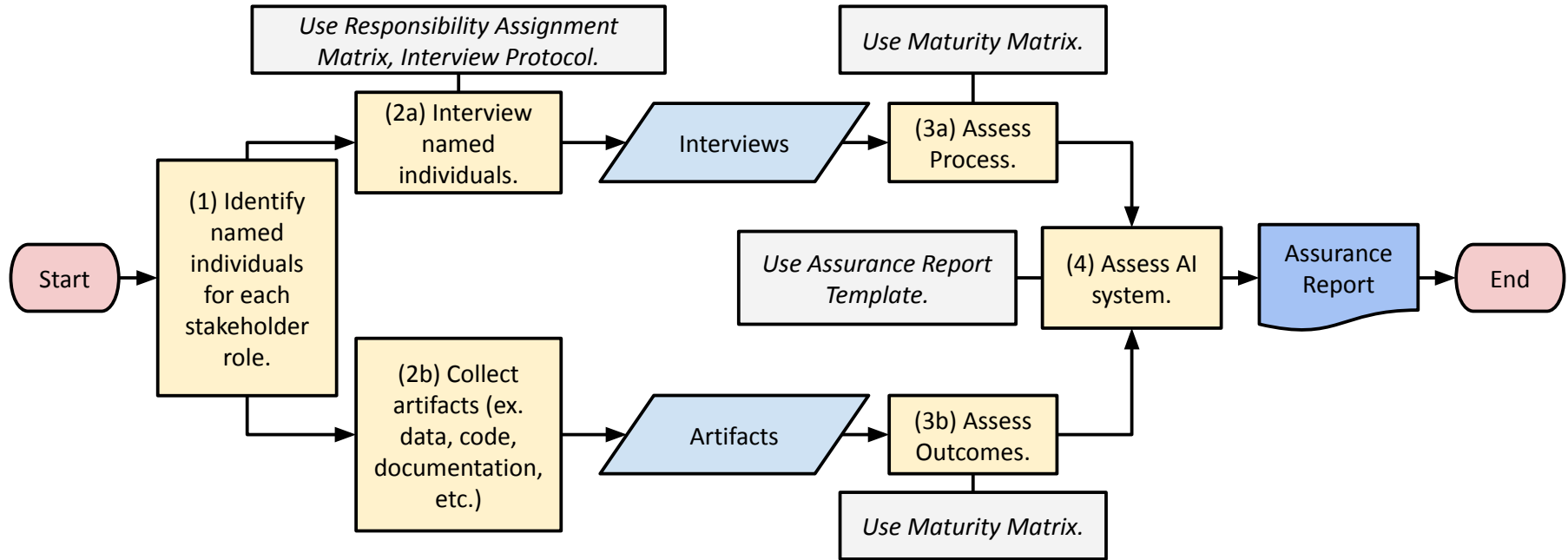


Design Requirements

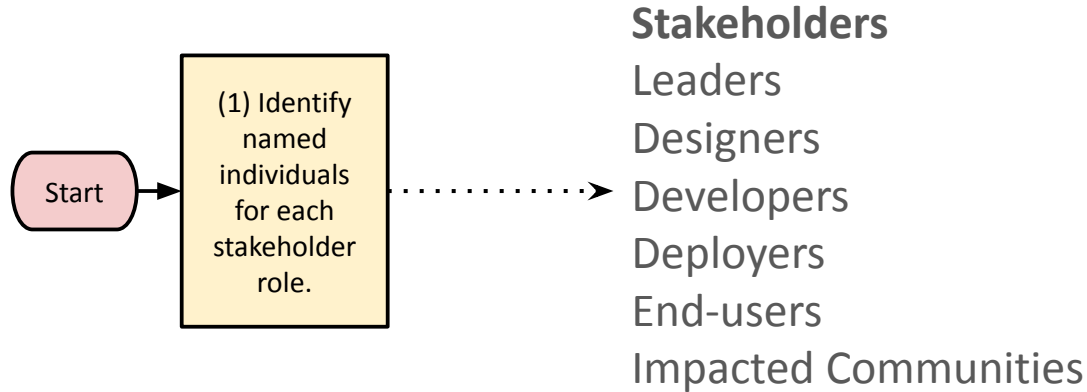
We identify five design requirements for an AI assurance process from prior literature.

- R1: Focuses on both process and outcome
- R2: End-to-end – across AI system stakeholders and AI lifecycle stages
- R3 : Offers clear instructions and guidance to conduct assurance
- R4: Offers concrete recommendations for improvement
- R5: Tested with real-world systems

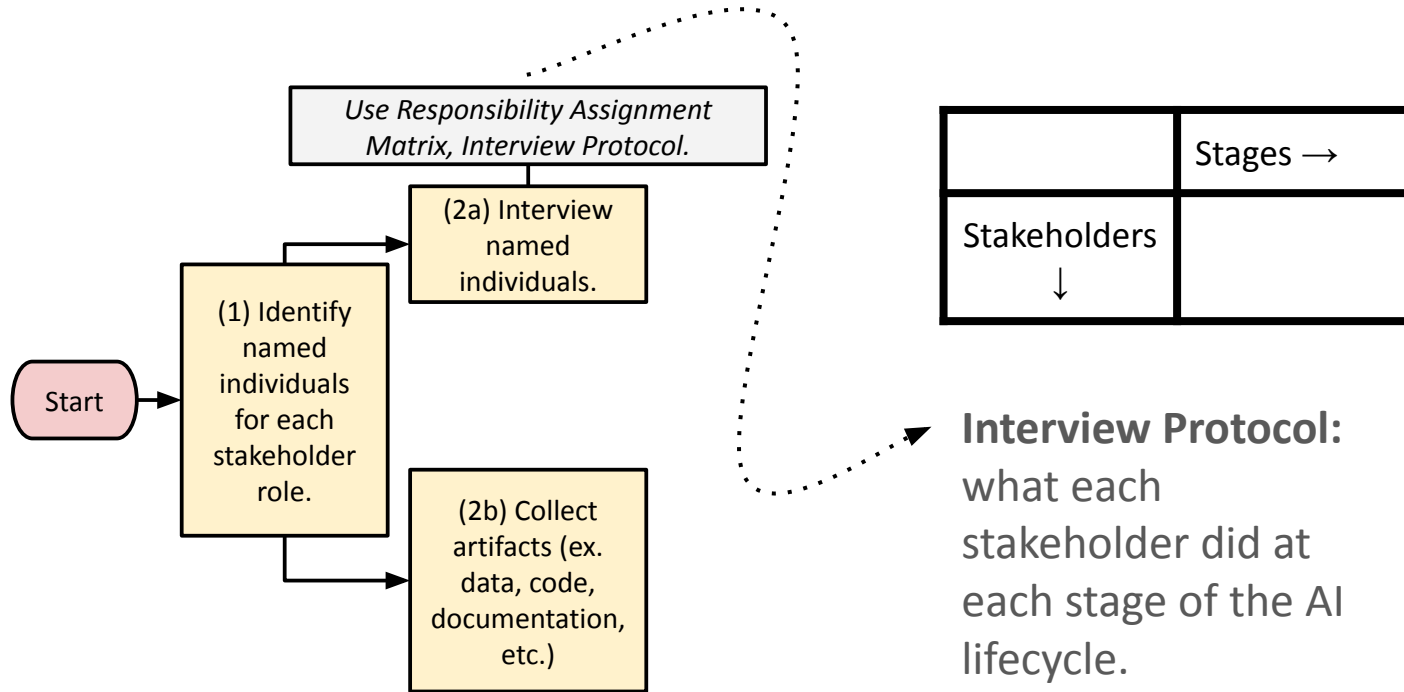
Prototype: AI Assurance Framework



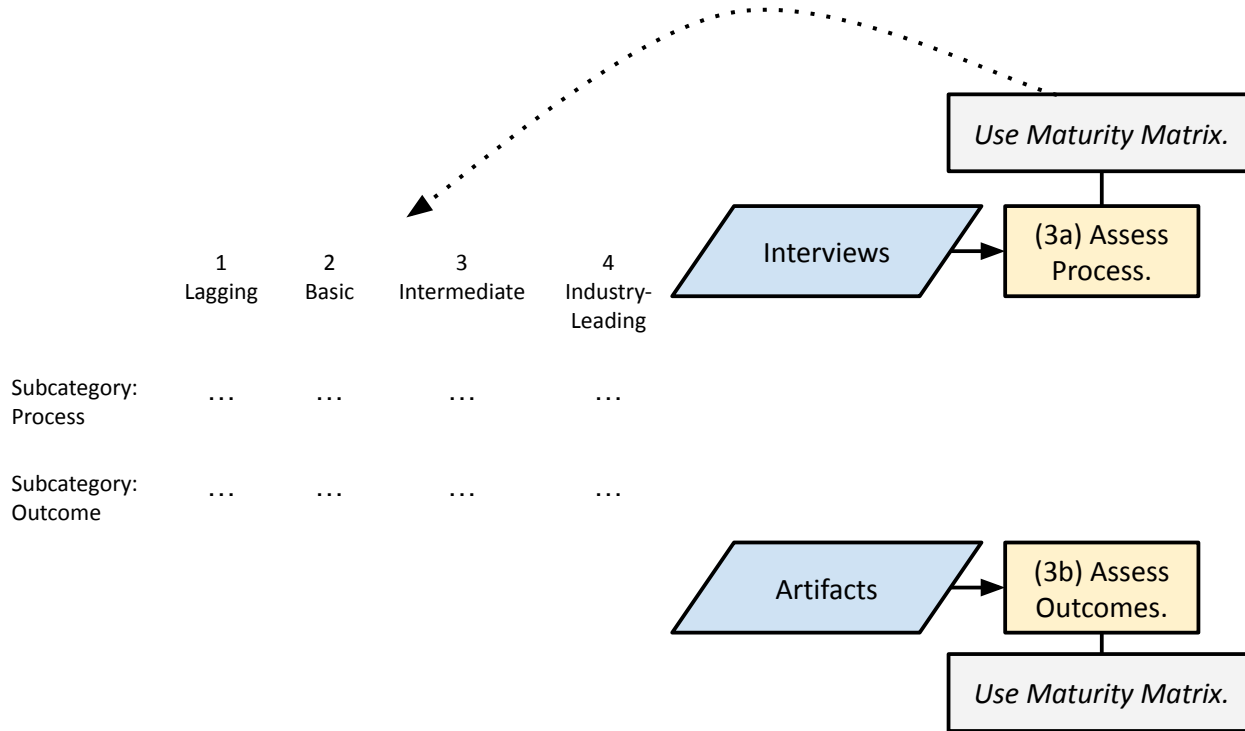
Prototype: Step (1)



Prototype: Steps (2a) and (2b)



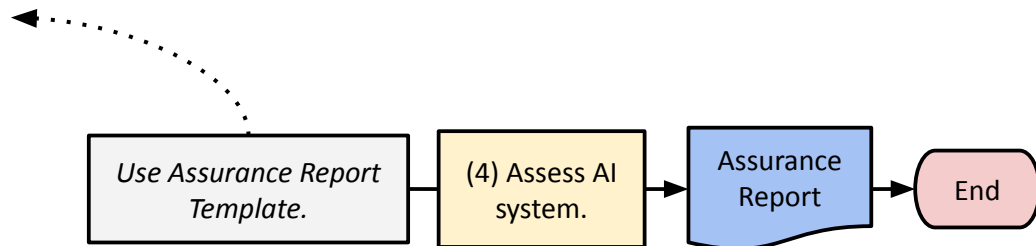
Prototype: Steps (3a) and (3b)



Prototype: Step (4)

Proposed Grades

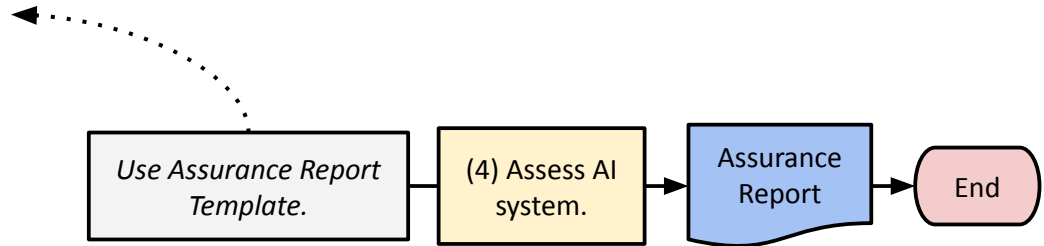
1. Inadequate access to perform the evaluation.
2. Adequate access, lack of compliance with baseline industry expectations.
3. Adequate access; adequate compliance with baseline expert expectations but lack of evidence of compliance with state-of-the-art and best practices.
4. Adequate access, adequate compliance with best practices, and state-of-the-art industry standards.



Prototype: Step (4)

Assurance Report

- Purpose
- Scope
- Opinion on AI system (1-4)
- Basis of opinion
- Table with AI system's grade on each subcategory
- Recommendations



Early Testing: Methodology

Pilot Applications

We apply our AI assurance framework to two real-world AI systems designed, developed, and deployed in different organizational settings.

- Business document tagging tool for downstream processing in a large private firm
 - 4 stakeholder interviews: leaders, designers, developers, deployers.
- Housing resource allocation in a public-sector agency
 - 7 stakeholder interviews: leaders, designers, developers, deployers, end-users.

Expert Validation Interviews

We conduct interviews with 6 experts in AI assurance, auditing, and / or governance.

We provided a 10-minute presentation of our framework, and ask about the framework's:

- Necessity
- Soundness
- Usability
- Effectiveness

Early Testing: Findings on Our Framework's Necessity

- Participants pointed to **the need for a standardized third-party assurance process**.
- Both stakeholders and experts highlighted a lack of standards in current practices: governance is mostly **internal, ad-hoc**, and **use case specific**
- Our framework is “*very helpful*” because “*it promotes [...] a mutual understanding of what good looks like*” (Expert 1)

Early Testing: Findings on Our Framework's Soundness

- Most experts consider the framework sound
 - **End-to-end** across both stakeholders and stages
 - **Definitions** of the stakeholders and stages **that reflect real-world systems**
 - **Separation between process and outcome** is reasonable
 - **Maturity matrix grades reflect today's practices**

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 - **Separation between process and outcome** is reasonable
 - **Maturity matrix grades reflect today's practices**
- However, applications of our framework should address:
 - The **lack of clear role assignments in practice**
 - The **iterative nature of AI development**
 - **Skipping certain stages** of the AI lifecycle (with the rise of **third-party models**)

Early Testing: Findings on Our Framework's Usability

- The framework is **accessible** and **easy to learn**
- Participants identified several considerations surrounding the **applicability** of our framework
 - *“I theoretically get [the assurance framework], but it’s so far [...] outside the scope of [...] how I’m used to thinking about things that I don’t know if I was helpful.” (S9)*
 - Some experts saw the framework as **too high-level**

Early Testing: Findings on Our Framework's Effectiveness

- The framework can identify problems in AI systems
 - E.g. the business document tagging model was built as a result of a top-down push from leadership
- **Conducting assurance in the early stages of the AI lifecycle is important!**
 - Enables **early detection of problems** when they are **easier and less costly to address**
 - Addresses **lost and changing information**

Early Testing: Caveats

Our validations are exploratory. No causal claims should be made from them.

- We were not able to conduct outcomes assessment for the pilot applications
- The pilot applications were both retrospective examinations of existing systems
- We did not compare whether AI systems improved using our assurance framework vs. status quo mechanisms for governance

Summary

- We design a **third-party AI assurance framework** that addresses our five design requirements by following well-established principles of design
- We conduct **early testing** of this framework in two ways
 - **Pilot applications** to two real-world AI systems
 - **Expert validation interviews**
- We find that overall, our framework is necessary, sound, usable, and effective
- We find that **flexibility in the assurance process** and **creating incentives to undertake assurance** (e.g. by conducting assurance in the early stages of the AI lifecycle) are important future directions

Thank you! Any questions?

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Prototype: Steps (3a) and (3b): Concrete Example

	1 Lagging	2 Basic	3 Intermediate	4 Industry-Leading
Organizational Feasibility: Process	Did not consider how task is done today. Did not consider how to embed AI system into existing business processes.	Considered how task is done today. Took some steps to understand best practices or the business model.	Considered how task is done today. Has approval on how AI system will fit into existing business processes or require a change. Knowledge of existing processes could be supplemented by ex. asking more people.	Considered how task is done today. Has approval on how the AI system will fit into existing business processes or require a change in business processes.
Organizational Feasibility: Outcome	Status quo better than AI system across all relevant organizational feasibility dimensions.	AI system somewhat aligns with existing best practices. Requires some major changes to workflows; with these, the AI system does not fit well with business model.	AI system mostly aligns with existing best practices. Requires some major changes to workflows; with these, the AI system fits into the business model.	AI system aligns with existing best practices and fits into business model.